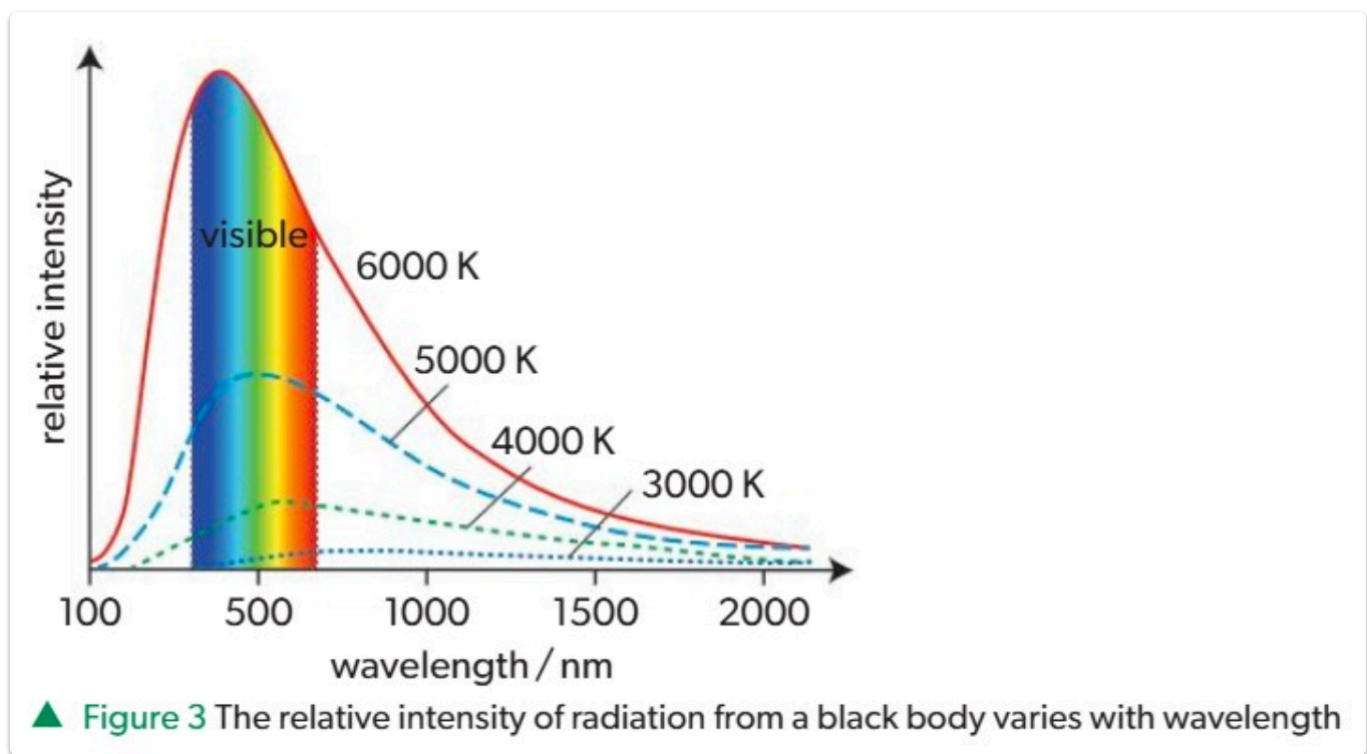


greenhouse effect

Radiation that arrives at the Earth's surface is **black-body radiation** from the Sun, which is considered a **black-body**.

Thermal radiation is emitted as electromagnetic waves by **all objects** at temperatures greater than 0 Kelvin. EM waves travel through a vacuum.

A **black body** absorbs all the radiation incident on it. A black body also emits radiation with a pattern characterised by its temperature; it is **black-body** radiation.



Intensity I of radiation is the power P emitted per unit area;

$$I = \frac{P}{A}$$

where A is the surface area of the emitting object.

The area under [this curve](#) is given by the **Stefan-Boltzmann law**, which predicts that the **intensity output by a black-body radiator** is $I = \sigma T^4$ where σ is the Stefan-Boltzmann constant.

In astrophysics, P is known as the **luminosity** L of a perfect **black body**:

$$I = \frac{P}{A} \equiv \frac{L}{A}$$

$$\sigma T^4 = \frac{L}{A} \quad \therefore \text{Stefan-Boltzmann law}$$

$$\therefore L = A\sigma T^4$$

For a grey body, $L = eA\sigma T^4$ where e is the emissivity of the body.

Wien's displacement law predicts the characteristic peak in [this graph](#) is given as:

$$\lambda_{\max} T = 2.9 \times 10^{-3} \text{ mK}$$

IMPORTANT: the mK in the equation above means '**metres kelvin**', *not milli-kelvin* or something else.

Apparent brightness b is an astronomical term used for the **amount of light** from a star or other body that **reaches Earth**. It is the **intensity** of the radiation at the orbit of the Earth.

b has the units W m^{-2} .

The **apparent brightness** of the Sun is known as the solar constant S . It is the **power that is incident** on **one square metre** at the top of the Earth's atmosphere.

The energy per second transferred to Earth from the Sun is the **overall difference** between the **incoming solar radiation** and the radiation that Earth **re-emits** back into space.

However, the energy from the Sun falls on only **half** of the area of the Earth's sphere at any one time. The energy is then transferred to the **whole** of Earth's surface.

Assuming that the Earth is spherical, it has a surface area of $4\pi R^2$ — but of course, everyone knows what shape the Earth actually is.

So, the average incident intensity I_{surface} during 24 hours at any point on the surface must be:

$$I_{\text{surface}} = \frac{S \times \pi R^2}{4\pi R^2} = \frac{S}{4}$$

$\therefore I_{\text{surface}}$ has a maximum value of $0.25 \times 1360 = 340 \text{ W}$.

The actual mean power that is incident on each square metre of the Earth's surface is **lower** than the calculated value of 340 W because the Sun's energy arriving at the surface is **reduced** due to:

- Radiation is absorbed and (Rayleigh **ray-lee**) scattered by the atmosphere
- The radiation passes through a greater thickness of atmosphere at dawn and dusk than in the middle of the day $\implies$ more scattering and absorption take place when the thickness of the atmosphere is greater.
- Extremely in-depth information [here](#).



Rayleigh scattering causes my favourite gradient to occur :)

The theoretical **black body** cannot be realised in practice. Real objects are known as grey bodies—their radiated emission is **less** than that of an equivalent black body at the **same temperature**. The extent to which an emitter is imperfect compared with a black body is described by **emissivity**, e .

$$\text{Emissivity} = \frac{\text{energy radiated from the surface of an object}}{\text{energy radiated from a black body at the same temperature}}$$

$$e = \frac{\text{power radiated per unit area from the surface of an object}}{\sigma T^4}$$

Hence,

$$e = \begin{cases} 1 & \text{for a black body} \\ 0 & \text{for a perfectly reflecting and non-radiating object} \\ 0 < x \in \mathbb{R} < 1 & \text{otherwise} \end{cases}$$

e has no units because it is a ratio of energies.

The Earth's surface is a **grey body** and scatters some of the incident energy back into the **atmosphere**. The extent to which the surface does this is known as its albedo.

$$\begin{aligned} \text{Albedo} &= \frac{\text{energy scattered by a given surface in a given time}}{\text{total energy incident on the surface in the same time}} \\ &= \frac{\text{total scattered power}}{\text{total incident power}} \end{aligned}$$

Different expressions using Albedo can be formed.

Let $\alpha = \text{Albedo}$.

Let $L_{scat} = \text{total scattered power} = \text{total reflected power}$

Let $L_{inci} = \text{total incident power}$

Let $L_{radi} = \text{total radiated power}$

$$\begin{aligned} \frac{L_{scat} + L_{radi}}{L_{inci}} &= 1 \\ \frac{L_{scat}}{L_{inci}} &= \alpha \\ \therefore \frac{L_{radi}}{L_{inci}} &= 1 - \alpha \\ L_{radi} &= (1 - \alpha)L_{inci} \end{aligned}$$

The above derivation has to be **memorised**. The formula in the book of depression is only **partial**, including L_{scat} and L_{inci} and α only.

Note that, in some texts, $L_{\odot}$ means the luminosity of the sun

Greenhouse gases to memorise:

- Water vapour (H_2O)
- Carbon dioxide (CO_2)
- Methane (CH_4)
- Nitrous oxide (dinitrogen monoxide) (N_2O)

Remember: **N**ot **M**y **C**all, **W**alter (breaking bad heheheheheheeeeee)

Worked Questions I found difficult

A planet orbits at a distance d from a star. The power emitted by the star is P . The total surface area of the planet is A .

Explain why the power incident on the planet is $\frac{P}{4\pi d^2} = \frac{A}{4}$.

Answer (unfold):

By imagining a sphere with radius d with a centre at the same position as the centre of the star, the intensity of the star at distance d can be obtained. And since $I = P \div A$, the A in $P = AI$ would be the cross-sectional area of the planet which is $\frac{A}{4}$. This is due to the inverse square law.